

1 Ablation Study

We conducted ablation experiments to evaluate the impact of different modality features and the contrastive learning strategy on model performance. Specifically, we ablated the relevant network structures, namely contrastive learning (CL), cross-modal contrastive learning between sequence and graph modalities (CL_{12}), between graph and structural modalities (CL_{23}), and between sequence and structural modalities (CL_{13}). As shown in Table 1, the proposed cross-modal contrastive learning strategy enhances TriMod-DTI’s performance, achieving absolute improvements of 1.1% in AUC and 2.0% in AUPR compared to the non-contrastive baseline. Furthermore, compared to the results of "w/o CL_{12} ", "w/o CL_{13} ", and "w/o CL_{23} ", TriMod-DTI’s cross-modal contrastive learning strategy among the three modalities demonstrates superior performance. The models "w/o 1D", "w/o 2D", and "w/o 3D" denote the removal of sequence features, graph features, and structural features, respectively. We conducted these three sets of experiments to evaluate the contribution of each modality to the overall performance. The results presented in Table 1 show that, regardless of which two modalities are used to represent drugs and proteins, TriMod-DTI outperforms these models in all three evaluation metrics. This indicates that integrating all three modalities (i.e., sequence, graph, and 3D structural features) enables the model to learn more complex and diverse drug-target representations, thereby enhancing its predictive power.

Table 1. Ablation study on GPCR datasets.

Method	AUC	AUPR	Precision
w/o 1D	0.847 ± 0.002	0.838 ± 0.005	0.747 ± 0.011
w/o 2D	0.837 ± 0.008	0.838 ± 0.002	0.722 ± 0.028
w/o 3D	0.858 ± 0.001	0.846 ± 0.003	0.746 ± 0.022
w/o CL	0.848 ± 0.010	0.859 ± 0.015	0.750 ± 0.012
w/o CL_{12}	0.864 ± 0.006	0.859 ± 0.007	0.749 ± 0.002
w/o CL_{23}	0.865 ± 0.005	0.862 ± 0.009	0.748 ± 0.010
w/o CL_{13}	0.868 ± 0.003	0.867 ± 0.006	0.746 ± 0.008
TriMod-DTI	0.870 ± 0.015	0.868 ± 0.002	0.751 ± 0.010

2 Parameter Sensitivity Analysis

In this section, we systematically investigated four key parameters in the TriMod-DTI model, including dropout rate, learning rate, number of GCN layers, and number of attention heads in the cross-attention layer. We conducted a comprehensive sensitivity analysis of these parameters on both the GPCR and Human datasets. Specifically, we evaluated the performance of dropout rates within the range of $\{0.1, 0.2, 0.3, 0.4\}$ and tested learning rates within the range of $\{1e-5, 1e-4, 1e-3, 1e-2\}$. Additionally, we adjusted the number of GCN

layers for both drugs and proteins, evaluating configurations of $\{1, 2, 3, 4\}$ layers, as well as the number of attention heads in the cross-attention layer within the range of $\{2, 4, 8, 16\}$.

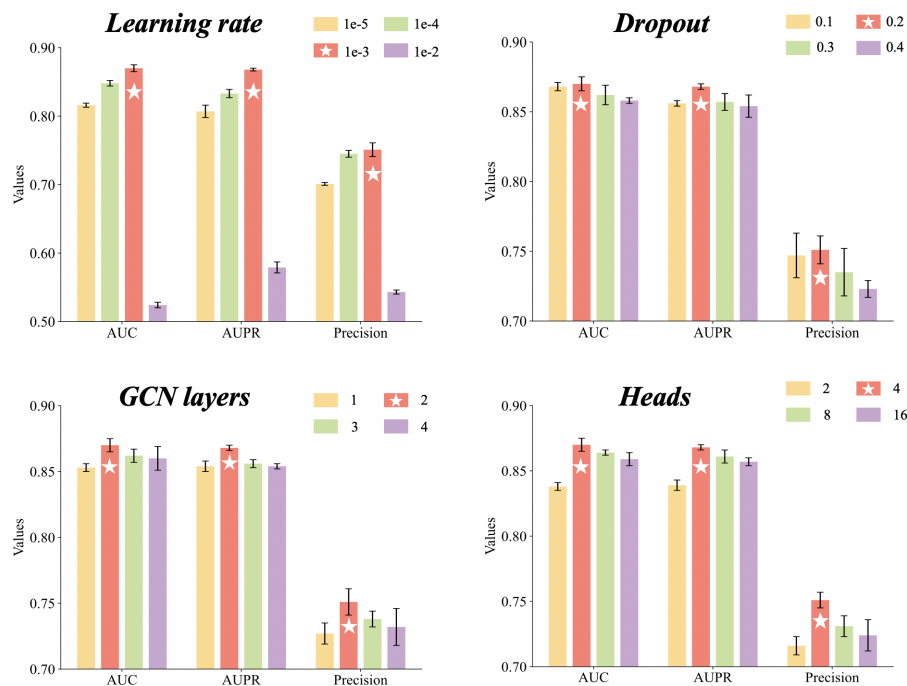


Fig. 1. Parameter sensitivity analysis for TriMod-DTI on GPCR dataset.

The parameters used by TriMod-DTI are marked with ★ in the figure. As shown in Figure 1, on the GPCR dataset, the model achieved optimal performance when the dropout rate was set to 0.2, the learning rate to 1e-3, the number of GCN layers to 2, and the number of attention heads to 4. Figure 2 further illustrates the experimental results on the Human dataset, indicating that the model also attained the best predictive performance when the dropout rate was 0.2, the number of GCN layers was 2, and the number of attention heads was 4. However, unlike the GPCR dataset, the Human dataset reached its optimal performance at a learning rate of 1e-4. Through a comparative analysis of the parameter trends between the GPCR and Human datasets, we observed that the AUC and AUPR metrics of the Human dataset were relatively less sensitive to variations in dropout rate, number of GCN layers, and number of attention heads, demonstrating stronger robustness.

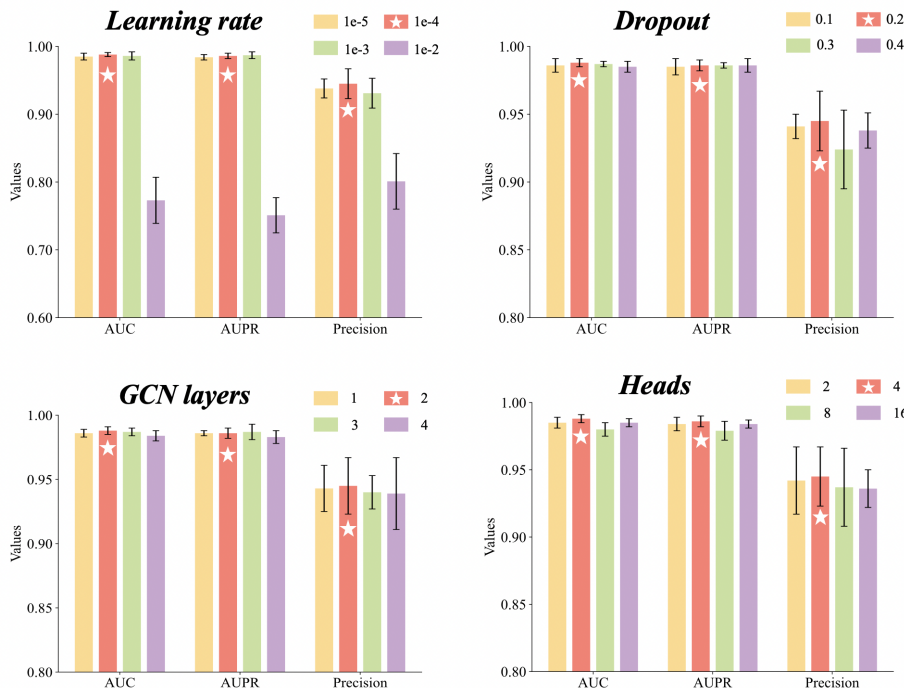


Fig. 2. Parameter sensitivity analysis for TriMod-DTI on Human dataset.

3 Case Study

Predicted Candidate Targets. As shown in Table 2, 5 out of the 10 drug-target pairs listed in the table were successfully predicted (highlighted in bold). Taking the drug *Verapamil* (DrugBank ID: DB00661) and its candidate target *Glucose-6-phosphate isomerase 2* (Uniprot ID: P13376) as an example, the target plays a crucial role in cellular metabolism, where it is responsible for generating NADPH in the pentose phosphate pathway, thereby maintaining intracellular redox balance. The activity of *Glucose-6-phosphate isomerase 2* is essential for cells to resist oxidative stress and sustain survival. On the other hand, *Verapamil* is a calcium channel blocker, and literature suggests that it significantly reduces the activity of *Glucose-6-phosphate isomerase 2* by modulating calcium signaling pathways and redox balance, thereby enhancing the cytotoxic effects of the mTORC1 inhibitor Everolimus on T-cell acute lymphoblastic leukemia cells.

Predicted Candidate Drugs. Table 3 shows the candidate predicted drugs for the *D(2) dopamine receptor* (Uniprot ID: P14416), with 5 out of the 10 predictions supported by evidence found in PubMed. These cases demonstrate that TriMod-DTI can effectively predict potential drug-target pairs, providing robust support for accelerating drug discovery and development.

Table 2. The predicted candidate targets for drug *Verapamil*.

Rank	Target Name	Target Uniprot ID	Evidence
1	Meso-diaminopimelate D-dehydrogenase	P04964	Unconfirmed
2	Adenylate kinase	P69441	Unconfirmed
3	Glucose-6-phosphate isomerase 2	P13376	PMID: 36978873
4	Purine nucleoside phosphorylase	P00491	Unconfirmed
5	Heparin cofactor 2	P05546	PMID: 31713254
6	Proteasome subunit alpha type-5	P28066	PMID: 27813130
7	Ras-related protein Ral-A	P11233	Unconfirmed
8	Cytochrome b6-f complex subunit 8	P83798	PMID: 37831396
9	Ornithine decarboxylase antizyme 1	P54368	PMID: 2491756
10	Dihydropteridine reductase	P09417	Unconfirmed

Table 3. The predicted candidate drugs for target *D(2) dopamine receptor*.

Rank	Drug Name	DrugBank ID	Evidence
1	Molybdenum	DB11137	PMID: 31196984
2	Apadenoson	DB05009	Unconfirmed
3	4-Carboxyphenylboronic Acid	DB03140	PMID: 32803981
4	Rilzabrutinib	DB17709	Unconfirmed
5	Glycyltyrosine	DB15991	Unconfirmed
6	Solamargine	DB12700	PMID: 16099621
7	Liothyronine	DB00279	PMID: 34986373
8	Cilnidipine	DB09232	Unconfirmed
9	N-hydroxyguanidine	DB03770	PMID: 15044095
10	Cobalt hexammine ion	DB03350	Unconfirmed